

Chapter: 26

ATOMIC PHYSICS

ATOMIC SPECTRA:

Definition:

The branch of Physics deals with the spectrum of frequencies of electromagnetic radiation emitted or absorbed by an electron during transitions between different levels of energy within an atom called Atomic Spectra.

Explanation:

These spectra occur because the electrons within an atom are confined to discrete and specific energy states. When an atom undergoes energy changes, its electrons transition between these specific levels, resulting in the absorption or emission of a photon with a particular frequency or wavelength. This specific frequency manifests as a **spectral line**.

Because these transitions happen at discrete intervals, atoms in a gas produce **line spectra**, rather than continuous bands of light.

These line spectra can be observed in two main ways:

1. **Emission spectra:** The specific wavelengths of radiation given off by radiating bodies or excited atoms.
2. **Absorption spectra:** A series of dark lines superimposed on a continuous spectrum, which is created when white light passes through a gas or dilute solution and the atoms absorb specific frequencies of the light.

Example:

A familiar example of atomic spectrum is the rainbow; though it appears to be a continuous band of colors, it actually contains faint black lines that represent the absorption spectra of the sun.

OPTICAL SPECTRA:

Definition:

An optical spectrum is obtained when light passes through a prism or a diffraction grating. This spectrum can be obtained from the intensity and wavelength of the radiation.

Explanation:

The spectrum may be observed visually in the limited wavelength region to which the eye is sensitive; it may be focused on a photographic plate.

1. EMISSION SPECTRA:

The spectra obtained from radiating bodies are called emission spectra.

These are classified into three types:

1. Continuous spectra:

These are emitted by solids, liquids and dense opaque gases at high temperatures.

The spectrum of the sun, or of a black body, is a continuous spectrum. Gases at low pressures emit line spectra.

2. Band spectra:

These are associated with similar changes in the molecules under high pressure.

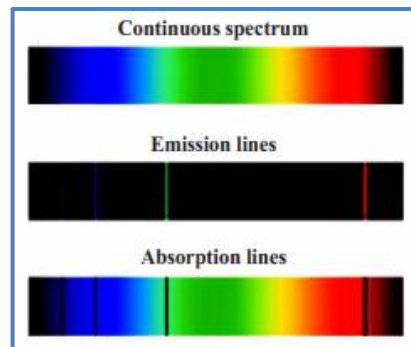
3. Line spectra:

These have their origin in the energy changes which take place in the atoms of a gas, while

2. ABSORPTION SPECTRA:

An absorption spectrum is used to analyze the substance.

This spectrum is obtained by passing white light from a continuous source through a gas or a dilute solution of the element being analyzed.



The absorption spectrum consists of a series of dark lines superimposed on the continuous spectrum of the light source.

SPECTRAL LINES:

Definition:

A line of particular frequency or wavelength emitted or absorbed by atoms is called spectral line.

Explanation:

When the electron in a hydrogen atom jumps between quantized energy levels, it emits or absorbs photons, creating grouped series of spectral lines.

The wavelengths are calculated using the **Rydberg formula**:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{p^2} - \frac{1}{n^2} \right)$$

Where R_H is the Rydberg constant $1.097 \times 10^7 \text{ m}^{-1}$.

The different series of hydrogen spectrum include:

1. **Lyman Series:** Jumps to/from ground state ($p = 1$).

$$\frac{1}{\lambda} = R_H \left(\frac{1}{1^2} - \frac{1}{n^2} \right), n = 2, 3, 4, \dots$$

2. **Balmer Series:** Jumps to/from $p = 2$.

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right); n = 3, 4, 5, \dots$$

It contains four visible lines: red, green, blue, violet at wavelengths 656 nm, 486 nm, 434 nm, and 410 nm.

3. **Paschen Series:** Jumps to/from $p = 3$.

$$\frac{1}{\lambda} = R_H \left(\frac{1}{3^2} - \frac{1}{n^2} \right).$$

4. **Brackett Series:** Jumps to/from $p = 4$.

$$\frac{1}{\lambda} = R_H \left(\frac{1}{4^2} - \frac{1}{n^2} \right).$$

5. **Pfund Series:** Jumps to/from $p = 5$.

$$\frac{1}{\lambda} = R_H \left(\frac{1}{5^2} - \frac{1}{n^2} \right).$$

Where R_H is a constant called the Rydberg constant.

THE SPECTRUM OF HYDROGEN ATOM:

The shape of the spectrum proposed by Max Planck in 1900, that the energy of a given mode of vibration depended upon its frequency in accordance with the relationship

$$E_n = nh\nu; \quad (n = 1, 2, 3, \dots)$$

This relation shows that the energy associated with each frequency for different transitions has a discrete value.

IDENTIFICATION OF ELEMENTS BY UNIQUENESS OF SPECTRA:

Each element has its own unique line spectrum which is referred to as the fingerprint of a particular element.

The spectrum for each element is unique because each element contains different numbers of electrons and different energy levels. The absorption spectrum of an element has many practical applications.

For example, various absorption lines observed in the solar spectrum have been used to identify elements in the solar atmosphere.

BOHR'S MODEL AND ITS POSTULATES:**Introduction:**

In 1913, Niels Bohr introduced atomic model in order to give quantitative determination of frequency emitted during de-excitation of an electron in Hydrogen-atom.

The discovery of the electron in the late 19th century marked the beginning of a new era in scientific research, helping physicists understand the structure and nature of atoms.

Niels Bohr had a remarkable idea that explained the atom of hydrogen-consist of a proton and an electron revolving around the nucleus.

POSTULATES OF BOHR'S ATOMIC MODEL:**Postulate: 1**

An electron bound to the nucleus in an atom can move around the nucleus in certain circular orbits without radiating energy. These shells or orbits are called the energy levels or discrete stationary states of the atom.

Postulate: 2

Angular momentum of an electron in its orbit or shell is constant and is equal to the integral multiple of $\frac{h}{2\pi}$

i.e. $mvr = \frac{nh}{2\pi}$

Where $n = 1, 2, 3, \dots$ and 'n' is called principal quantum number.

m = mass of the orbiting electron

v = velocity of the orbiting electron

And h = Plank's constant

Postulate: 3

Whenever an electron jumps from higher energy state to a lower energy state, the excess energy is emitted as a photon, so that $E_n - E_p = h\nu$

Where E_n = higher energy level

E_p = lower energy level

h = Plank's constant

And $\nu = \frac{c}{\lambda}$ is the frequency of the radiation emitted.

BOHR'S RADIUS:

According to Bohr's postulates, the hydrogen atom consists of nucleus containing a proton only and only one electron revolves around the nucleus in definite circular orbit.

Since the charge of nucleus is positive (due to one proton) and the charge of circulating electron (around the nucleus in a definite orbit) is negative. Therefore, electrostatic force of attraction on electron, due to proton, towards nucleus is

$$F_e = K \frac{e^2}{r^2} \dots \dots \dots (1)$$

Where each of the two particles (electron and proton) has a charge “e” and “K” is Coulomb’s constant and is equal to $9 \times 10^9 \text{ N.m}^2\text{C}^{-2}$.

The electron revolves in a circular orbit of radius “r” with the velocity “v”, this velocity provides a centrifugal force to the electrons as follows:

$$F_c = \frac{mv^2}{r} \dots \dots \dots (2)$$

Since Coulomb’s attractive force is balanced by the centrifugal force. Therefore

$$\begin{aligned} F_e &= F_c \\ \frac{Ke^2}{r^2} &= \frac{mv^2}{r} \\ \frac{r}{r^2} &= \frac{mv^2}{ke^2} \\ \frac{1}{r} &= \frac{mv^2}{Ke^2} \\ r &= \frac{ke^2}{mv^2} \dots \dots \dots (3) \end{aligned}$$

For the velocity of electrons, Bohr used his postulate of angular momentum of electron.

$$mvr = \frac{nh}{2\pi}$$

Or

$$v = \frac{nh}{2\pi rm}$$

Squaring both sides, we have

$$v^2 = \frac{n^2 h^2}{4\pi^2 r^2 m^2}$$

Putting the values of v^2 in equation (3), we get

$$\begin{aligned} \therefore r &= \frac{Ke^2}{m(n^2 h^2 / 4\pi^2 r^2 m^2)} \\ r &= Ke^2 \times \frac{4\pi^2 r^2 m}{n^2 h^2} \\ \frac{r}{r^2} &= \frac{4\pi^2 Kme^2}{n^2 h^2} \\ r &= \frac{n^2 h^2}{4\pi^2 Kme^2} \end{aligned}$$

Where, $k = \frac{1}{4\pi\epsilon_0}$. Therefore

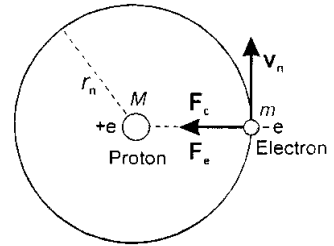
$$\begin{aligned} r &= \frac{n^2 h^2}{4\pi^2 k(\frac{1}{4\pi\epsilon_0})me^2} \\ r &= \frac{n^2 \epsilon_0 h^2}{\pi me^2} \\ r &= \left(\frac{\epsilon_0 h^2}{\pi me^2} \right) n^2 \\ r &= r_0 n^2 \end{aligned}$$

Here $h = 6.63 \times 10^{-34} \text{ Js}$, $m = 9.1 \times 10^{-31} \text{ kg}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$.

Therefore, $r_0 = 5.29 \times 10^{-11} \text{ m} = 0.53 \text{ \AA}$ is called Bohr's first orbit.

As the values of r is depending only on the principal quantum number n . Therefore, the radius has quantized values.

$$r_n = r_1 n^2$$



ORBITAL ENERGY OF HYDROGEN ATOM:

The allowed energy levels and quantitative values of the emission wavelengths of the hydrogen atom can be calculated from the postulate (3) suggests the qualitative existence of a characteristic discrete emission spectrum and corresponding absorption spectrum for hydrogen.

The electron, revolving around the nucleus does not radiate energy, as given by classical electromagnetic theory but instead it is assumed that the total energy of the atom remains constant.

The total energy of an atom is the sum of kinetic energy and potential energy of electron i.e.

$$E_n = P.E + K.E$$

1. KINETIC ENERGY OF ELECTRON:

It is represented by K.E, and according to formula, it is

$$K.E = \frac{1}{2} m v^2 \dots\dots\dots (7)$$

Centrifugal force, F_c of electron is equal to electrostatic force or centrifugal " F_e " on electron, therefore

$$\begin{aligned} F_c &= F_e \\ \frac{m v^2}{r} &= \frac{K e^2}{r^2} \\ m v^2 &= \frac{K e^2}{r} \end{aligned}$$

By putting the values of $m v^2$ in equation (7), we get

$$K.E = \frac{K.e^2}{2r} \dots\dots\dots (8)$$

2. POTENTIAL ENERGY OF ELECTRON:

Since,

P.E = Work done on a body against a force.

Or

P.E = (Negative Force) (Distance)

$$= \left(-\frac{K e^2}{r^2} \right) (r)$$

Or

$$P.E = \frac{-K e^2}{r}$$

TOTAL ENERGY OF ATOM:

$$E_n = P.E + K.E$$

$$= \frac{-K e^2}{r} + \frac{K e^2}{2r}$$

$$= \frac{-2K e^2 + K e^2}{2r}$$

$$E_n = \frac{-K e^2}{2r} \dots\dots\dots (8)$$

By putting the values of, $r = \frac{n^2 h^2}{4\pi^2 K m e^2}$ in equation (8)

We get,

$$E_n = \frac{-K e^2}{2 \left(\frac{n^2 h^2}{4\pi^2 K m e^2} \right)}$$

$$E_n = -K e^2 \times \frac{2\pi^2 K m e^2}{n^2 h^2}$$

$$E_n = -\frac{2\pi^2 K^2 m e^4}{n^2 h^2}$$

Or

$$E_n = -\frac{m e^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2} \right)$$

$$E_n = -(E_0) \frac{1}{n^2}$$

$$E_n = \frac{-E_0}{n^2} \dots\dots\dots (9)$$

Where E_0 is constant and is equal to 13.6 eV, it is called ionization energy to remove an electron from Bohr's atom.

$$E_0 = \frac{2\pi^2 k^2 m e^4}{h^2} = \text{constant}$$

By putting the values of various elements, we get

$$E_0 = \frac{2\pi^2(9.1 \times 10^{-31}) \times (9 \times 10^9)^2 \times (1.6 \times 10^{-19})^4}{6.6 \times 10^{-34}}$$

$$= 2.17 \times 10^{-18} J$$

$$= \frac{2.17 \times 10^{-18}}{1.6 \times 10^{-19}} eV$$

$$\therefore E_0 = 13.6 eV$$

Hence, equation (9) becomes

$$E_n = -\frac{13.6}{n^2} eV \dots \dots \dots (10)$$

The negative sign shows that energy must be supplied from some external source to remove an electron from the atom or from lower energy orbit to higher energy orbit.

For $n = 1$, equation (10) becomes, $E_1 = -13.6 eV$

Which is the energy required to remove an electron from the first Bohr orbit. This is called ionization energy.

EXCITATION ENERGY:

"The energy needed to take the atom from its ground state to an excited state is called the excitation energy of that excited state". OR

"The energy required to excite (lift the electron from its ground state to higher energy orbit) an atom is called its excitation energy."

This energy is usually given to the atom by making a collision with accelerated electron from outside the atom. The process of transferring energy to an atom or molecule, raising it to a higher energy state, often resulting in the emission of photons as it returns to its ground state.

IONIZATION ENERGY:

"The minimum energy required to remove an electron from an atom or molecule, resulting in the formation of an ion, is called Ionization Energy". OR

"The energy required to completely remove an electron from an atom is called ionization energy."

The energy levels corresponding to various shells " n " is given in equation.

$$E_n = -\frac{13.6}{n^2} eV$$

The lowest level for $n = 1$ is the ground state of hydrogen atom and higher levels correspond to excited states. The energy for the greatest value of $n = \infty$ become zero and for any energy $E = 0$ in which the electron and proton do not bound together is called ionization energy.

Therefore, the energy $E = 13.6 eV$ is required to ionize the hydrogen atom.

THE SPECTRUM OF HYDROGEN ATOM:

The electron in a hydrogen atom can jump between quantized energy levels by emitting or absorbing Photon for some different values of wavelengths. Any such wavelength is often called a line spectrum which can be absorption or emission lines.

The lines for hydrogen are said to be grouped into series, according to the level at which upward jumps start and downward jumps end.

The formula for these series corresponding to the different wavelengths can be obtained from equation

$$E = -\frac{me^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2} \right)$$

The various different wavelengths of electromagnetic radiations are emitted from hydrogen atom; the collection of all these different wave lengths is called "emission spectrum". This spectrum consists of different series:

1. Lyman series 2. Balmer series 3. Paschen series

4. Bracket series 5. Pfund series

In order to explain emission of these series, Bohr's theory was introduced.

Suppose that the electron in the hydrogen atom is in the higher state 'n' with energy 'E_n' and makes a transition to a lower state 'p' with energy 'E_p'. Then according to the postulate of Bohr's theory

$$\Delta E = E_n - E_p \dots\dots\dots (1)$$

Here

$$\Delta E = h\nu$$

$$E_n = \frac{-E_0}{n^2}$$

And

$$E_p = \frac{-E_0}{p^2}$$

Putting the values of E_n and E_p in equation (1) we get,

$$h\nu = -E_0 \left(\frac{1}{n^2} - \frac{1}{p^2} \right)$$

$$\frac{hc}{\lambda} = -E_0 \left(\frac{1}{n^2} - \frac{1}{p^2} \right)$$

Since $\nu = \frac{c}{\lambda}$. Therefore

$$\frac{1}{\lambda} = \frac{-E_0}{hc} \left(\frac{1}{n^2} - \frac{1}{p^2} \right)$$

Or

$$\frac{1}{\lambda} = \frac{E_0}{hc} \left(\frac{1}{p^2} - \frac{1}{n^2} \right)$$

Or

$$\frac{1}{\lambda} = \frac{me^4}{8\varepsilon_0^2 h^3} \left(\frac{1}{p^2} - \frac{1}{n^2} \right)$$

Or

$$\frac{1}{\lambda} = R_H \left(\frac{1}{p^2} - \frac{1}{n^2} \right)$$

Where $R_H = \frac{me^4}{8\varepsilon_0^2 h^3} = 1.097 \times 10^7 \text{ m}^{-1}$ is called Rydberg constant.

The frequency of the photon emitted when the electron makes a transition from an orbit to an inner orbit as stated in postulate: 3.

$$\nu = \frac{E_n - E_p}{h}$$

OR

$$\nu = \frac{me^4}{8\varepsilon_0^2 h^3} \left(\frac{1}{p^2} - \frac{1}{n^2} \right)$$

To express above equation in terms of wavelength, it is convenient to use $c = \nu\lambda$

The possible series associated with the different wavelengths can be explained from the above equation.

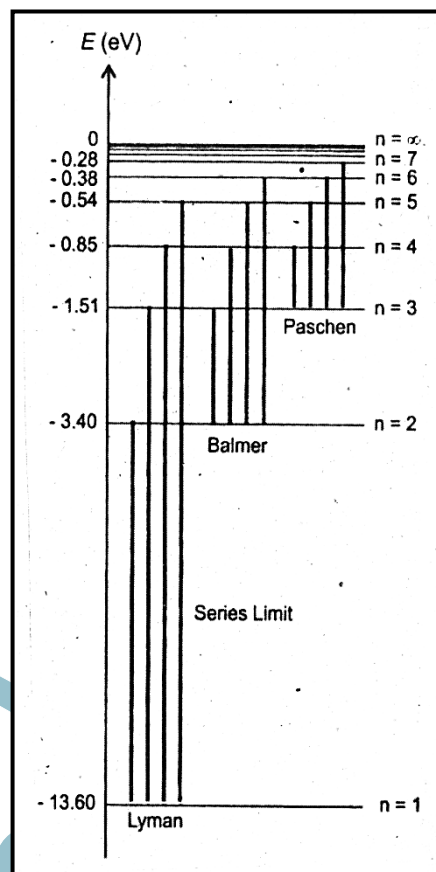
The emission and absorption lines for all possible jumps up from the $n = 1$ level and down to the $n = 1$ level are said to be in the Lyman series. Similarly, other series are illustrated in the figure.

WORKED EXAMPLE: 26.1

The electron in a hydrogen atom makes a transition from the $n = 3$ energy level to the ground level ($n = 1$). Find the wavelength and frequency of the emitted photon.

WORKED EXAMPLE: 26.2

The lowest energy levels of a mercury atom are shown below. The diagram is not to scaled. Calculate the frequency of an emitted photon due to a transition, shown by an arrow, from level $n = 4$ to level $n = 3$. Which transition would cause the emission of a photon of a longer wavelength than that emitted in the transition from level $n = 4$ to level $n = 3$?



X-RAYS:

INNER SHELL TRANSITIONS:

ELECTRON TRANSITION:

The passage of electrons from one energy level to another is called electron transition.

Explanation:

When the transition of electrons from the outer shell to the inner shell occurred then the electromagnetic radiation corresponded to the characteristic energy generated as shown in figure.

The transition of electrons in "H" or lighter elements results in the emission of spectral lines in the infrared, visible or UV regions of electromagnetic spectrum due to small energy differences in transition levels.

In heavy atoms electrons are arranged in concentric shells K, L, M, N, O, etc.

Electrons in the outer shell are loosely bound with nucleus, so a small amount of energy is required to excite electrons from outer shell and when it returns to original state, it emits low energy photon.

On the other hand, inner shell electrons are tightly bound electrons, so a large amount of energy is required to excite it and when this electron returns to their ground state, they emit high energy photons.

These high energy radiations are named as x-rays or characteristic x-rays.

X-RAYS:

Introduction:

In 1895, X-rays were accidentally discovered by a German scientist, Dr. Roentgen while studying cathode rays in discharge tube. He observed another type of radiation that could be detected outside the tube, and it was originally called Roentgen rays.

Roentgen could not find out whether the radiation was a stream of particles or a train of waves, he decided to call it X-rays, "x" stands for unknown.

Roentgen found that some photographic plates, kept carefully wrapped in his laboratory, had become fogged. Instead of merely throwing them aside he set out to find the cause of the fogging.

He traced it to a gas-discharge tube, which he was using with low pressure and high voltage. This tube appeared to emit radiation that could penetrate paper, wood, glass, rubber, and even aluminum a centimeter and a half thick.

Definition:

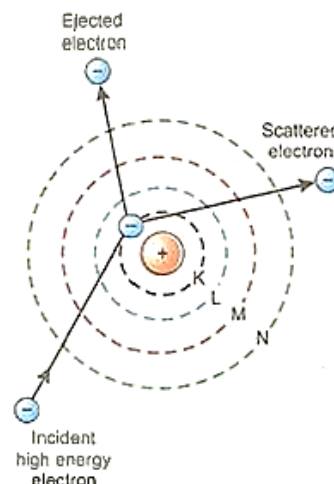
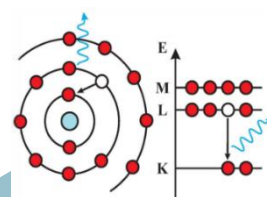
"When fast moving electrons strike on a metal surface, photons of high frequency or energy are emitted; these emitted photons are called X-rays."

X-Rays are electromagnetic waves having wavelengths in the range of 0.01nm to 10nm.

Explanation:

The production of X-rays is a reverse process of photoelectric effect. X-rays are produced when a target metal is bombarded by electrons; while in photo electric effect, the electrons are produced when light waves are incident on a metal surface.

X-rays are powerful electromagnetic rays of higher energy and can pass through most objects, including the body. Most of them have a wavelength ranging from 0.01 to 10 nanometers, corresponding to frequencies 3×10^{19} Hz to 3×10^{16} Hz.

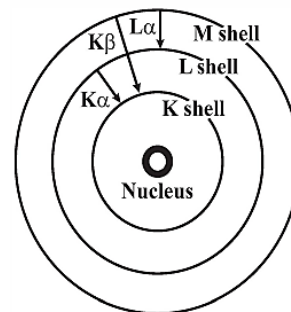


In an X-ray tube, very energetic electrons are bombarded on the atoms in a metal target, the atoms will be left in an excited state with a hole in the electron shell (As in figure).

When this hole is filled by an electron from an outer shell, an x-ray photon with energy equal to the difference in the electron energy levels is produced. The energy of the x-ray photon is characteristic of the target metal.

The different characteristic lines occurred due to the electron transitions between different energy levels as shown in figure. Electron transitions into an atom, which produce the K_{α} , K_{β} and L_{α} characteristic x-rays.

X-rays are used to generate images of tissues and structures inside the body.



PRODUCTION AND CHARACTERISTICS OF X-RAYS:

Construction:

A modern X-rays unit consists of the following parts.

1. A high vacuum tube of glass or metal, called X-rays tube.
2. A source of electrons, that is cathode plate (C) with Filament (F).
3. A metal target "T", usually called tungsten disc which serves as anode (A) on which the electrons strike to produce X-rays.
4. A supply of about 10 volts (L.T) for heating the filament.
5. Another high voltage supply (V) is about 100 kV (H.T) for accelerating electrons towards target.
6. A Colling system behind metal target.

Working:

When the cathode plate "C" is heated by a white-hot tungsten filament "F", with a low voltage battery called low tension (L.T), it emits electrons by thermionic emission which are accelerated towards the anode "T".

In the figure, F is the filament (Cathode) and T is the target (Anode).

If V is the potential difference between the cathode "C" and the target "T", which is applied by the high voltage battery called (H.T). The kinetic energy with which the electrons strike the target is given by

$$\text{K.E.} = Ve \dots \dots \dots (1)$$

Suppose that these fast-moving electrons of energy " Ve " strike the target "T" made of heavy metal such as cadmium or any other heavy metal. It may be possible that in a collision, the electrons in the innermost shells, such as K or L will be knocked out.

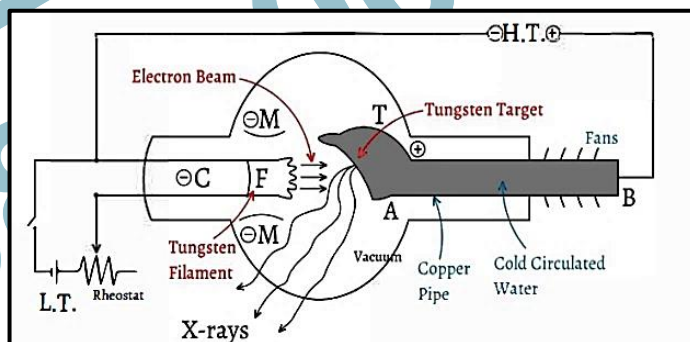
Because there is so little gas, the electrons on their way to the anode do not lose any noticeable amount of energy in ionizing atoms.

During the production of X-rays 99% of energy of electron converted to heat and 1% into x-rays photon. The heat generated at the target by the electronic bombardment is so enormous that the target must be cooled artificially.

X-RAYS SPECTRUM:

The emitted X-rays spectrum consists of two parts.

1. Characteristic x-rays.
2. Continuous x-rays.



1. CHARACTERISTIC X-RAYS SPECTRA:

The part of X-rays consists of series of peaks (K_α , K_β , K_γ) is known as characteristic x-rays.

Definition:

The photons emitted in the inner shell transition are called characteristics X-rays, because their energies depend upon the type of target material.

The holes created in the L and M shells are occupied by transition of electrons from higher states creating more X-rays. The characteristics X-rays appear as discrete lines on a continuous spectrum.

Production of Characteristic x-rays:

When a fast-moving electron strikes with heavy element like tungsten, the electron from the inner most shells, such as 'K' or 'L' may knock out.

If one electron from K-shell is removed, thereby producing a vacancy or hole in K-shell. The electron from L-shell fills this vacancy by emitting a photon of energy ' $hf_{K\alpha}$ ' called ' K_α ' x-ray

$$hf_{K\alpha} = E_L - E_K \dots\dots\dots (1)$$

The electron from 'M' shell may also fill vacancy by emitting a photon of energy ' $hf_{K\beta}$ ' called ' K_β ' x-rays.

$$hf_{K\beta} = E_M - E_K \dots\dots\dots (2)$$

So, K_α , K_β , K_γ , results from transition into K-shell from L, M, N, Shells respectively.

When L_α , L_β , results from transition into L-shell from M, N, Shells respectively.

2. CONTINUOUS X-RAYS SPECTRA:

The part of X-rays consists of a continuous wavelength of x-ray known as Bremsstrahlung x-rays.

Definition:

"The X-rays which are emitted in all directions and with a continuous range of frequencies are called continuous X-rays."

The continuous X-rays spectrum is obtained due to slowing down (deceleration) of impacting electrons.

Production of Continuous x-rays:

The continuous spectrum takes place due to an effect known as Bremsstrahlung or braking radiation.

When the fast-moving electrons are bombarded on the target, they are suddenly slowed down in impact with the target. We know that an accelerating charge emits electromagnetic radiation. Hence, these impacting electrons emit radiation as they are decelerated by the target.

Since the rate of deceleration is so large, the emitted radiation corresponds to short wavelength and so the bremsstrahlung is in the X-ray region.

Energy of emitted photons is equal to the difference of K.E of electrons.

$$hf = (K.E)_i - (K.E)_f$$

Since the rate of deceleration is so large, the emitted radiation corresponds to short wavelength and so the bremsstrahlung is in the X-ray region.

When the electrons lose all their Kinetic Energy, in the first collision, the entire K.E appears as X-rays photons of energy $h \cdot f_{\max}$ i.e.

$$h \cdot f_{\max} = (K.E)_{\max} \dots\dots\dots (1)$$

But the Kinetic energy of electrons is

$$K.E = eV \dots\dots\dots (2)$$

By comparing equation (1) and equation (2), we get

$$e \cdot V = hf$$

$$e \cdot V = \frac{hf}{\lambda}$$

$$\therefore f = \frac{c}{\lambda}$$

$$\lambda = \frac{hc}{e.V}$$

$$\text{OR } \lambda_{min} = \frac{hc}{e.V}$$

The wavelength λ_{min} corresponds to the maximum frequency f_{max} .

Other electrons do not lose all their energy in the first collision. They may suffer a number of collisions before coming to rest. This will give rise to photons of smaller energy or X-rays of longer wavelength.

Hence, the continuous spectrum is produced due to deceleration of impacting electrons.

PROPERTIES OF X-RAYS:

1. They travel in straight lines with velocity of light ($3 \times 10^8 \text{ ms}^{-1}$)
2. They are not deflected by electric and magnetic fields. This shows that they are neutral.
3. They can be diffracted by crystals; this property is used in crystallography.
4. They cause ionization in gases, and this property is used to find out the intensity of X-rays.
5. They can produce fluorescence in many substances like zinc sulphide, barium platino cyanide or cadmium tungstate by exciting their atoms.
6. X-rays can cause photoelectric emission on striking some metal surface.
7. They can damage living cells (tissues).
8. X-rays are highly penetrating radiation. They can pass through even those substances which are opaque to ordinary light i.e. they can pass through wood and flesh.
9. X-rays are a part of electromagnetic spectrum with frequency higher than that of ultra-violet rays.
10. Compton Effect can be observed with them.

USES OF X-RAYS:

1. Higher energy X-rays are used to destroy cancer cells within the body.
2. X-rays are used at customs and security posts to detect arms, explosive materials and contraband goods.
3. X-rays are widely used in industry. For example, X-rays photographs show hidden flaws such as cracks, air holes or imperfections in welds.
4. They are mostly used in diagnostics of bone fractures.
5. X-rays have many practical applications in medicine and industry. Because X-rays can penetrate several centimeters into a solid matter so they can be used to visualize the interiors of the material such as dislocation of bones or defects in structural steel.

LASER

LASER: LIGHT AMPLIFICATION BY STIMULATED EMISSION OF RADIATION:

Laser is the abbreviation of light amplification by stimulated emission of radiation.

Definition:

It is a device which is used to produce intense, monochromatic & unidirectional coherent beam of visible light.

Explanation:

The electromagnetic waves experienced in daily life, ranging from the sun, stars, incandescent and fluorescent lamps are emitted from atoms or molecules spontaneously.

- Ordinarily natural and artificial light is released by energy changes on the atomic and molecular level that occur without any outside interference.
- A second type of light exists, when an atom or molecule retains its excess energy until stimulated or induced to emit the energy in the form of light.

Lasers are designed to produce and amplify this stimulated form of light into intense and focused beams.

Compared to conventional sources of ordinary light, the light from a laser is quite intense, monochromatic, and emitted in a unidirectional beam limited by diffraction.

The special nature of laser light has made laser technology a vital tool in nearly every aspect of everyday life including communications, entertainment, manufacturing, and medicine.

WORKING PRINCIPLE:

The excited atoms are stimulated by incident photons in order to emit another photon of the same frequency as incident photons.

To understand working we should know the following terms.

1. Stimulated absorption.
2. Spontaneous emission.
3. Stimulated emission.
4. Population inversion.
5. Meta-stable state.

1. STIMULATED OR INDUCED ABSORPTION:

Definition:

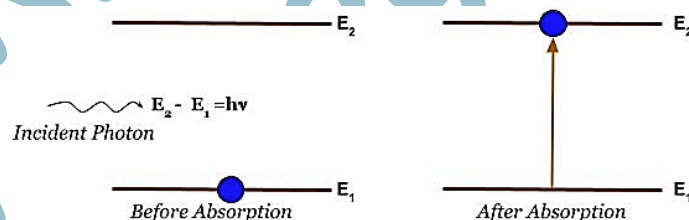
The process in which optical energy is converted to internal energy of atom is called Stimulated Absorption.

This is called so because the incident photon has stimulated the atom to absorb energy.

Explanation:

When a photon of light having energy ($E_2 - E_1 = h\nu$) is incident on an atom in the ground state, the atom in the ground state E_1 may absorb the photon and jump to a higher energy state E_2 . This process is called stimulated absorption or induced absorption.

The lifetime of an atom in an excited state is 10^{-8} seconds.



2. SPONTANEOUS EMISSION:

Definition:

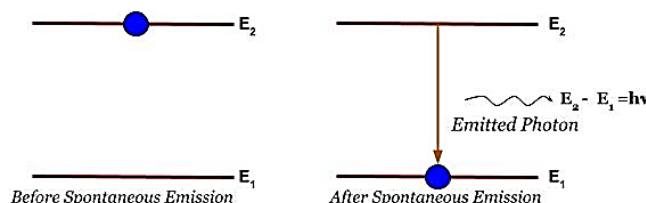
The process of photon emission by an excited atom with no external influences is called spontaneous emission. OR

The process in which an atom in the excited state makes a transition to a lower state spontaneously with emission of a photon is called spontaneous emission.

Explanation:

Since in excited state the lifetime of an atom is too short therefore, a probability that the atom in excited state E_2 will go back to the lower state E_1 by spontaneously emitting the absorbed radiation ($E = h\nu$) is very high.

The emitted radiation is incoherent to the induced energy photon.



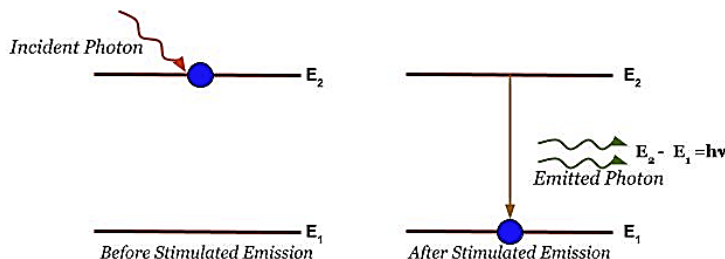
3. STIMULATED EMISSION:

Definition:

When an excited atom's electron interacts with a photon, it decays to an available lower energy level, and in the process gives off another photon, such a process is called Stimulated Emission.

Explanation:

In this process an atom is initially in an excited state, a passing photon of energy equal to the difference of two energy levels induces the atom to emit a photon and make a transition to lower level.



In the stimulated emission two photons are involved:

1. The primary photon, causing the emission of radiation by an excited atom and
2. The secondary photon emitted by the atom.

These two identical photons will be exactly in phase and coherent.

If a stimulated or induced emission occurs, the number of identical photons increases, the light amplification in the laser. The beam is coherent because all photons are in phase, the beam is monochromatic because the photons all have the same wavelength, and the beam is coherent because all photons move in same direction and phase.

4. POPULATION INVERSION:

Population inversion is key to producing laser light.

Definition:

The process of achieving a greater population in a higher energy state (Excited state) compared to a lower energy state (ground state).

Explanation:

The number of electrons per unit volume in an energy state is the population of that energy state.

Consider a group of electrons with two energy levels E_1 and E_2 .

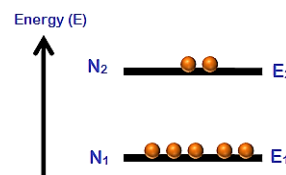
- E_1 is the lower energy state (Ground state) and
- E_2 is the higher energy state (Excited state).
- N_1 is the number of electrons in the energy state E_1 .
- N_2 is the number of electrons in the energy state E_2 .

A sample of free atoms in which more atoms are in the ground state than an excited state is known as normal population. In normal population of energy state

$$N_1 > N_2$$

When there are more atoms in the excited state than ground state, this is called an inverted population. In population inversion of energy state

$$N_2 > N_1$$



Population inversion causes more stimulated emission and can be achieved if there exists a metastable state. This is mainly used for light amplification, and it is required for laser operation.

5. METASTABLE STATES:

Definition:

The excited state in atoms in which electrons stay for a slightly longer time period before spontaneously falling to ground state is called metastable state.

Explanation:

A metastable state is an excited state in which an excited electron is unusually stable and from which the electron spontaneously falls to lower state only after relatively longer time. The transition from or to this state is difficult compared to other excited states.

Hence, instead of direct excitation to this state, the electrons are excited to higher level to spontaneously fall to metastable state.

Consider a system consisting of three energy levels E_1 , E_2 , E_3 with N number of electrons.

We assume that the energy level of E_1 is less than E_2 and E_3 , the energy level of E_2 is greater than E_1 and less than E_3 , and the energy level of E_3 is greater than E_1 and E_2 .

It can be simply written as.

$$E_1 < E_2 < E_3.$$

That means the energy level of E_2 lies in between E_1 and E_3 .

The energy level E_2 is sometimes referred to as Meta stable state having a lifetime of 10^{-3} seconds. This energy level E_2 makes stimulated emission possible.

6. LASER PUMPING:

The act of energy transfer from an external source, producing excited state in its atom called pumping.

There are two ways of Laser Pumping.

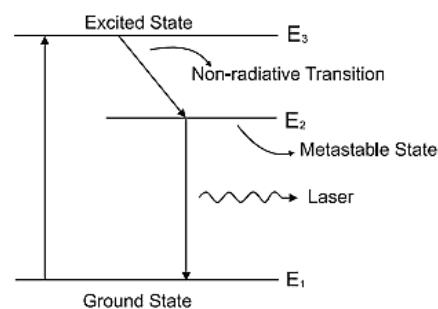
1. Pumping energy in the form of light is optical pumping.
2. Pumping energy in the form of current is electrical pumping.

LASER ACTION:

Consider a material whose atoms can reside in three different energy states E_1 , E_2 and E_3 .

Where:

- E_1 is ground state.
- E_3 is an exciting state in which the atoms can reside only for 10^{-8} sec and
- E_2 is a metastable state, in which the atoms can reside for 10^{-3} sec much longer than 10^{-8} sec.



Step: 1. Achieving Excited State by Optical Pumping:

When light (Optical) energy is supplied, which is equal to the energy difference between E_3 and E_1 , the electrons in the lower energy state (E_1) gain sufficient energy and jump into the higher energy state (E_3). This process of supplying energy is called optical pumping. The flow of electrons from E_1 to E_3 is called pump transition.

Step: 2. Achieving Metastable state by population inversion.

Since the lifetime of electrons in the energy state E_3 is very small as compared to the lifetime of electrons in the energy state E_2 . Therefore, electrons in energy level E_3 do not stay for long time period.

Because of the shorter lifetime, only a small number of electrons accumulate in the energy state E_3 .

After a short period, they quickly fall to the Meta stable state or energy state E_2 and releases radiation less energy instead of photons.

The electrons in the Meta stable state E_2 will remain there for a longer period because of their longer lifetime. As result, a large number of electrons accumulate in Meta stable state.

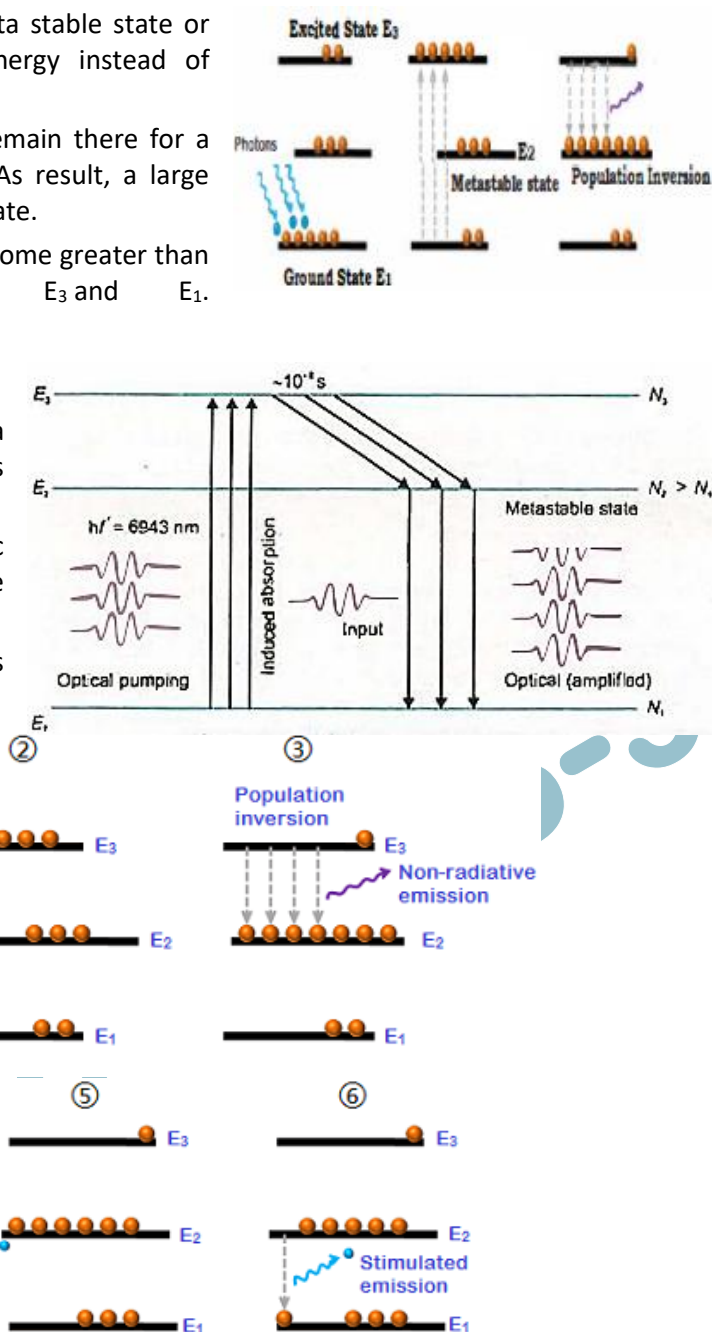
Thus, the population of metal stable state will become greater than the population of energy states E_3 and E_1 . It can be simply written as $N_2 > N_1 > N_3$.

Step: 3. For Stimulated Emission.

After the population inversion is achieved the Meta stable state E_2 is exposed to a beam of photons which causes induced (stimulated) emission.

The photons, so produced are monochromatic (same frequency) and coherent (in phase) in the direction of the incident photon.

The whole process for population inversion is shown in figure for three level systems.



Step: 4. Production of Laser.

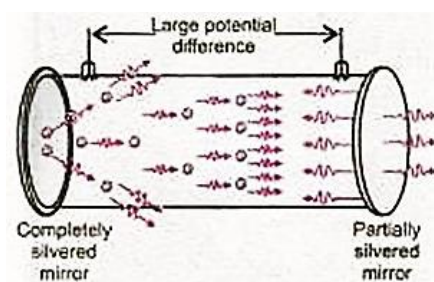
The emitted photons must be confined in the assembly long enough to stimulate further emission from other excited atoms.

This is achieved by using mirrors at the two ends of the assembly.

- One end is made totally reflecting, and
- The other end is partially transparent to allow the laser beam to escape.

As the photons move back and forth between the reflecting mirrors, they continue to stimulate other excited atoms to emit photons.

As the process continues the number of photons multiply, and the resulting radiation is, therefore, much more intense and coherent than light from ordinary sources.



HE-NE GAS LASER:***Introduction:***

The helium-neon laser was the first gas laser ever constructed, built in 1961 by Ali Javan, Bennett, and Herriott at Bell Telephone Laboratories.

Before this there was solid state laser called Ruby Laser. Ruby Laser emits light in pulses, but He-Ne-laser emits continuous waves.

Definition:

A gas laser in which a mixture of He (85%) & Ne (15%) gas is used as a laser medium at low pressure (1mm Hg) is called He-Ne Laser.

Explanation:

Helium-Neon laser is a type of gas laser in which a mixture of helium and neon gas is used as an active or gain medium and electrical pumping is used to achieve population inversion.

It is widely known for its coherent and monochromatic light output, typically emitting red light at a wavelength of 632.8 nm

To produce the laser beam, it is essential to achieve population inversion. Population inversion is the process of achieving more electrons in the higher energy state as compared to the lower energy state.

In general, the lower energy state has more electrons than the higher energy state. However, after achieving population inversion, more electrons will remain in the higher energy state than the lower energy state.

In order to achieve population inversion, we need to supply energy to the gain medium or active medium.

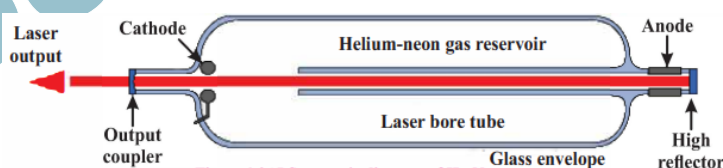
CONSTRUCTION:

It consists of a glass tube filled with helium and neon gases, electrodes to excite the gas, and mirrors at both ends to form an optical cavity.

The helium-neon laser consists of three essential components:

1. Pump Source: (High voltage power supply)

- The pump energy of the laser is provided by an electrical discharge of several hundred Volts between an anode and cathode at each end of the glass tube.
- A current of 5 to 100 mA is typical for laser operation.

***2. Laser Medium OR Gain Medium: (He & Ne)***

- A gas discharge tube contains a low-pressure mixture of helium-neon in a ratio between 5: 1 and 20: 1 bound in a glass tube.
- The partial pressure of helium is 1 mbar whereas that of neon is 0.1 mbar.
- Neon (Ne) is a lasing or active medium, and role of Helium (He) is just to excite Neon (Ne), whereas "Ne" is much heavier and could not be pumped efficiently without Helium.
- Helium and neon have nearly identical metastable states, respectively located 20.61 eV and 20.66 eV level.

3. Resonating Cavity (Laser glass tube or discharge glass tube):

- The Helium-neon gas laser system consists of a glass tube of nearly one meter in length.
- The glass tube (containing a mixture of helium and neon gas) is placed between two parallel mirrors. These two mirrors are silver or optically coated.

- Each mirror is silvered differently; the left side mirror is partially silvered and is known as an output coupler whereas the right-side mirror is fully silvered and is known as the high reflector or fully reflecting mirror.
- The fully silver mirror will completely reflect the light whereas the partially silver mirror will reflect most part of the light but allows some part of the light to produce the laser beam.

WORKING PRINCIPLE:

High voltage excites Helium atoms, which transfer energy to Neon atoms. This creates a population inversion, leading to stimulated emission of light.

WORKING OR LASER OPERATION:

1. Pumping:

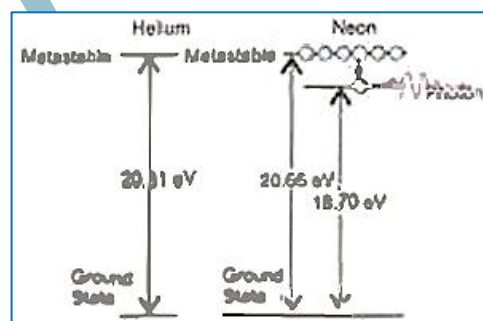
High voltage electric discharge excites electrons in Helium atom up to 20.61 eV energy state.

2. Population Inversion:

- Helium and Neon have nearly the same metastable states 20.61 eV and 20.66 eV respectively.
- Collision of the excited helium atoms with the ground-state neon atoms results in transfer of energy to the neon atoms, exciting neon electrons.
- The difference between the energy states of the two atoms (Helium 20.61 eV and Neon 20.66 eV) is in the order of 0.05 eV, which is supplied by kinetic energy.
- Therefore, excited Helium (He) atoms collide with Neon (Ne) atoms at ground state and transfer its 20.61 eV energy to 'Ne' atom along with 0.05 eV K.E.
- The number of neon atoms in the excited states builds up as further collisions between helium and neon atoms occur, causing a population inversion.
- As a result, electrons in Neon atom reside to 20.66 eV metastable state and population inversion is achieved.

3. Stimulated Emission:

- Spontaneous emission cause electron of Neon drops from metastable state energy 20.66 eV to ground state energy 18.70 eV with emission of a photon of energy 1.96 eV.
- This emitted photon moves back and forth between two mirrors in resonating cavity, cause stimulated emission and red-light photon ($\lambda = 632 \text{ nm}$) produced.



PROPERTIES AND USES OF LASERS:

PROPERTIES OF LASER:

The properties of laser light are very strange, as compared to conventional light. The factors that make laser light prominent are:

1. Monochromatic.
2. Highly Intense or Brightness
3. Coherence and
4. Directionality

1. Monochromatic:

A single color or wavelength light is called monochromatic.

The light emitted from ordinary light sources has different wavelengths, or colors. However, laser light has a single wavelength or color.

2. Highly Intense or Brightness:

The characteristics of coherence and directionality of the laser makes laser light highly intense as compared to conventional light.

Intensity is the power emitted per unit surface area per unit solid angle.

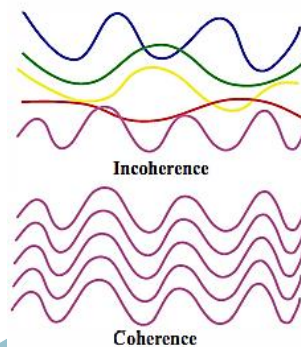
A one milliwatt He-Ne laser is more intense than the sun intensity.

3. Coherence:

Two or more waves of same frequency are said to be coherent in nature if they have constant phase difference.

A predictable correlation of the amplitude and phase at any one point with another point is called coherence.

In laser light the property of coherence occurs between any two or more light waves. Whereas in conventional light, the property of coherence exhibits between a source and its virtual source.



4. Directionality or Divergence:

The light ray coming from laser light travels in a single direction. However, an ordinary light source travels in all directions.

For example, on travelling 1 Km, the torch light spreads 1 km distance. But the laser light spreads only a few centimeters distance even if it travels lacks kilometer distance.

USES OF LASERS:

Lasers are widely used in many fields, some of them are:

1. Tools:

- Cutting tools are typically precise, and also simple to automate. They do not require sharpening, unlike that of knives.
- Robot-guided lasers are widely used for the cutting of pieces of cloth.
- Laser tools are used in corneal surgery, restoring a detached retina of eye, and removing kidney stones.

2. Communication:

- Lasers are used in Barcode scanners to convert printed barcode into a number.
- A semiconductor laser beam helps to convert the printed pattern of data into numbers in a CD/DVD.
- In Photonics, lasers are used in fibre optic cables.

3. Defense:

- In the defense field, the military employs laser-guided guns and missiles.
- Laser rangefinders use high-resolution scanning to find the distance and speed from an object that is located beyond the point-blank range.

4. Medicine:

- In medicine, the laser beam is used as part of phototherapy for many procedures.
- The development of laser technology in recent decades has enabled the creation of a new field of medicine, laser surgery.
- Lasers have uses in dermatology, ophthalmology, urology, rheumatology and dentistry.

5. Holography:

Holography is a true three-dimensional image recorded on film by lasers. Holograms are used for amusement, decoration on novelty items and magazine covers.

SAFE HANDLING OF LASERS:

Lasers come in various shapes and forms. They have many uses in teaching, research, manufacturing, medicine, dentistry, communications, and shop checkouts and most commonly at work in the office. In fact, some applications may be so well engineered that users are not even aware that the equipment contains a laser.

LASER BEAM HAZARDS:

The health effects that could occur due to exposure to a laser beam are damage to the skin and eyes. Skin effects include erythema, elastosis (photoaging), and immediate pigment darkening (tanning), burns and skin cancer.

Eye effects include photokeratitis, photo conjunctivitis, cataracts, photo retinal damage and burns. The nature of safety measures which are generally taken to avoid hazards, or damage can be caused by the lasers are adopted and procedures according to the use and intensity of the laser.

The following are some general safety protocols which must be taken into consideration while working with lasers.

- Only **trained, authorized personnel** may operate lasers, Authorization is received from the authorized laser user and the Laser Safety Officer.
- **NEVER** put yourself into any position where your eyes approach the axis of a laser beam (even with eye protection on).
- **Keep beam paths below or above** standing or sitting eye level. Do not direct them towards other people.
- **Do not damage laser protective housing or** malfunction the interlocks on these housings.
- **Eliminate all reflective material** from the vicinity of the beam paths.
- **Never use viewing instruments** to look directly into a laser beam. If this is necessary, install an appropriate filter into the optical element assembly.
- **Keep ambient light levels as high** as operations permit.
- **Do not work alone** when performing high power laser operations.

SUMMARY

- Atomic spectrum deals with the spectrum of frequencies of electromagnetic radiation emitted or absorbed by an electron during transitions between different levels of energy within an atom.
- An optical spectrum is obtained when light passes through a prism or a diffraction grating.
- A line of particular frequency or wavelength emitted or absorbed by atoms is called spectral line.
- The spectra for each element are unique because each element contains differing numbers of electrons and thus different energy levels.
- Niels Bohr explained the atom of hydrogen. Hydrogen atoms consist of protons and electrons revolving around the nucleus.
- According to Bohr's the radii of Hydrogen atom have the quantized values $r = \frac{n^2 E_0 h^2}{\pi m e^2}$ or $r = r_0 n^2$ where $r_0 = 5.29 \times 10^{-11} \text{m} = 0.53 \text{\AA}$ is called Bohr's first orbit.
- X-Rays are powerful electromagnetic rays of higher energy and can pass through most objects, including the body.
- X-rays are produced when fast electrons, or cathode rays, strike a target, such as the walls or anode of a low-pressure discharge tube.
- The light from a laser is quite intense, monochromatic, and one directional.
- The process of photon emission by an excited atom with no external influences is called spontaneous emission.
- Radiation which occurs as a result of external exposure is called induced or stimulated emission.

- Metastable state is a particular excited state of an atom, nucleus, or other system that has a longer lifetime than the ordinary excited states.
- The process of making the population of atoms in the higher energy state (Excited state) more than that of lower energy state (ground state) is known as population inversion.
- A process in which light is used to raise (pump) electrons from a lower energy level in an atom or molecule to a higher energy level is called optical pumping.
- The helium-neon laser was the first continuous wave (CW) laser ever constructed.
- Laser light is Monochromatic, Highly Intense, Coherence and single directional.
- Lasers are widely used in tools, communication, defense, medicine and holography.
- The health effects that could occur due to exposure to a laser beam are damage to the skin and eyes.
- The nature of safety measures which are generally taken to avoid the hazards or damage caused by the lasers are adopted and procedure according to the use and intensity of the laser.

TEXTBOOK EXERCISE

SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)

- Atomic spectra are also known as:
(a) Discrete spectra (b) Line spectra (c) Emission spectrum (d) Continuous spectra
- The ratio of kinetic energy to total energy for an electron in a Bohr orbit is:
(a) 1:1 (b) 2:3 (c) 1:2 (d) 2:1
- The Bohr radius increases as the principal quantum number:
(a) increases (b) decreases (c) remains constant (d) oscillates
- Laser beams consist of:
(a) Highly coherent electrons (b) Highly coherent photons
(c) Highly coherent phonons (d) Highly coherent neutrons
- The ruby laser is an example of:
(a) Optical pumping (b) Electrical pumping
(c) Chemical pumping (d) Thermal pumping
- Laser action requires a medium with at least: (2025)
(a) Three energy levels (b) Four energy levels
(c) Two energy levels (d) Five energy levels
- Population inversion occurs in:
(a) Active medium (b) Passive medium
(c) Gaseous medium (d) Vapour medium
- X-rays transfer ____ to metals.
(a) Energy (b) Force (c) Pressure (d) Momentum
- X-rays are deflected by:
(a) Magnetic fields (b) Electric fields (c) Gravitational fields (d) No fields
- Doubling the voltage of an X-ray tube:
(a) Halves the intensity (b) Keeps the intensity unchanged
(c) Doubles the intensity (d) Quadruples the intensity

SECTION- B: SHORT ANSWERED QUESTIONS

1. Why do different elements have different spectra?
2. In the Bohr model, how many times larger is the radius of the fifth Bohr orbit compared to that of the first Bohr orbit?
3. What is the difference between X-Rays and Gamma Rays?
4. State the properties of X-Rays, which makes it possible to detect cracks in bones.
5. What is the energy of a photon that, when absorbed by a hydrogen atom, could cause an electronic transition from (a) the $n = 3$ state to the $n = 5$ state and (b) The $n = 5$ state to the $n = 8$ state?
6. What are (a) wavelength range and (b) frequency range of the Lyman series and the Balmer series?
7. Distinguish between spontaneous and stimulated emission.
8. Explain why population inversion is necessary in a laser?
9. In an optically pumped laser, the light that causes optical pumping is always shorter in wavelength than the laser beam. Explain
10. A hydrogen atom is in its first excited state ($n = 2$). Calculate (a) the kinetic energy of the electron (b) the potential energy of the system, and (c) the total energy of the system.

SECTION- C: LONG ANSWERED QUESTIONS

1. What are the postulates of Bohr's Model of hydrogen atoms? Discuss the importance of this model to explain various series of line spectra in hydrogen atom. Do any of the assumptions of the Bohr's theory of hydrogen atom contradict with the classical Physics? Derive the expression for total energy of electron in n th Bohr orbit and show that: $E_n \propto 1/n^2$.
2. How are X-rays produced? State the purpose of cooling fins in the X-ray tube.
3. Explain why X-rays are appropriate in study of crystalline structure material? Write some main properties of X-rays.
4. What is Laser? Write the characteristics of Laser light. Can a two-level system be used for the production of Laser? Why?
5. What is pumping? What are the different methods of pumping? Explain optical pumping.
6. What is the principle of Laser? Write about the construction and working of Helium-neon laser.
7. Give some important properties of lasers. Also write about the uses of lasers in the field of medicine, defense and communication.
8. What is wave number? Derive the expression $\frac{1}{\lambda} = R_H \left(\frac{1}{p^2} - \frac{1}{n^2} \right)$
9. Derive the expression for Bohr's radius and develop a general relation for radii of quantized orbits of hydrogen atom.

SECTION- D: NUMERICAL

1. Calculate the energy of an electron in the $n = 2$ orbits of a hydrogen atom according to the Bohr model. **(Ans. -5. 447 x 10⁻¹⁹ J)**
2. Calculate the speed of the electron if it orbits in (a) the smallest allowed orbit and (b) the second smallest orbit? (c) If the electron moves to larger orbits, does its speed increase, decrease, or stay the same? **(Ans. (a) 2.19 x 10⁶ m/s (b) 1.09 x 10⁶ m/s (c) the speed of the electron will decrease)**
3. What are the (a) energy, (b) magnitude of the momentum, and (c) wavelength of the photon emitted when a hydrogen atom undergoes a transition from a state with $n = 3$ to a state with $n=1$? **(Ans. (a) 12.1 eV (b) 6.45 X 10⁻²⁷ N.s (c) 102 nm)**
4. What is the energy of the photon emitted by hydrogen atom when the hydrogen atom changes directly from the $n = 5$ state to the $n = 2$ state? **(Ans. 2.85 eV)**
5. How much work must be done to pull apart the electron and the proton that make up the hydrogen atom if the atom is initially in (a) its ground state and (b) the state with $n = 3$?

(Ans. (a) 13.6 eV (b) 1.51 eV)

6. (a) What is the wavelength of light for the least energetic photon emitted in the Balmer series of hydrogen atom spectrum lines? (b) What is the wavelength of the series limit? **(a) 659 nm (b) 366 nm)**
7. A laser emits light with a wavelength of 632.8 nm and has a power output of 55 mW. Calculate the energy of one photon emitted by this laser. **(Ans. 3.14×10^{-19} J)**
8. Calculate the wavelength of X-rays if the energy of one photon emitted by the X-ray machine is 1.9878×10^{-15} joules. **(Ans. 0.1 nm)** **(MP:25) (2025)**

ASSIGNMENT: 26

SECTION- A: MULTIPLE CHOICE QUESTIONS

Bohr's Atomic Model & Hydrogen Spectrum Series:

1. In 'L' shell of hydrogen atom, angular momentum of an electron is: **(2022)**
 (a) $\frac{h}{2\pi}$ (b) $\frac{2h}{\pi}$ (c) $\frac{h}{\pi}$ (d) $2\pi h$
2. The radius of the 1st orbit of hydrogen atom is: **(2021)**
 (a) 0.53m (b) **0.053nm** (c) 0.0053nm (d) 0.00053nm
3. The ratio of the radius of 3rd Bohr orbit in hydrogen atom to the radius of 1st Bohr orbit is: **(2019)**
 (a) 12:1 (b) 3:1 (c) 6:1 (d) 9:1
4. An electron can revolve in the nth orbit if, and only if, its angular momentum is equal to: **(2018)**
 (a) h (b) **nh** (c) $\frac{h}{\pi}$ (d) $\frac{nh}{2\pi}$
5. Lyman found a series of lines in the spectrum of hydrogen atom in this region of: **(2018)**
(a) Ultraviolet (b) Visible (c) infra-red (d) Far infra-red
6. Balmer series of Hydrogen atom spectrum lies in the: **(2016)**
 (a) Radio wave region (b) infrared region (c) **visible region** (d) ultraviolet region
7. Balmer series is obtained when the transitions of electrons terminate on: **(2015, 01)**
 (a) 1st orbit (b) **2nd orbit** (c) 3rd orbit (d) 4th orbit
8. When an electron falls from the 3 orbits to the 1 orbit in the Hydrogen atom, the lines spectrum obtained belongs to: **(2013)**
 (a) Bracket series (b) Lyman series (c) Balmer series (d) Paschen series
9. Brackett series of hydrogen atom spectrum lies in: **(2010)**
 (a) Ultraviolet region (b) Visible region (c) **Infrared region** (d) all of these
10. Hydrogen atom spectrum consists of lines in: **(2008)**
 (a) Region (b) visible region (c) infrared region (d) **all of them**
11. The frequencies in spectral lines emitted in Lyman series are in the _____ region: **(2006)**
 (a) Visible (b) Infrared (c) X-rays (d) **Ultraviolet**
12. According to Bohr's postulates, the electron revolving around nucleus in a fixed orbit radiates; **(2003)**
 (a) Energy (b) **No energy** (c) γ -rays (d) α -rays

Ionization & Excitation Energy:

13. The lifetime of the electron in excited state is: **(2014)**
 (a) 10^8 sec (b) **10^{-8} sec** (c) 10^{-3} sec (d) 10^3 sec

X- Rays:

14. To produce X-rays, an element of high atomic weight is bombarded by: **(2025)**
 (a) Photon (b) Electrons (c) Helium (d) neutron

15. X-rays are produced if heavier atoms are bombarded by: (2021)
 (a) Protons (b) **Electrons** (c) Neutrons (d) Positrons
16. The wavelength of X-ray is in the range: (2010)
 (a) 0.01 nm to 0.1 nm (b) $1\text{\AA}$ to $100\text{\AA}$ (c) 0.1 $\text{\AA}$ to 1m (d) **0.1 nm to 1.0 nm**
17. The fast-moving electrons stopped by a heavy metallic target in an evacuated glass tube, give rise to the production of: (2002M)
 (a) α -particle (b) β -particles (c) **X-rays** (d) Protons

Laser:

18. The least number of energy levels required for LASER action are: (2025)
 (a) Two (b) three (c) Four (d) Five
19. An intense, monochromatic, narrow and coherent beam of light is called: (2022)
 (a) X-rays (b) β -rays (c) LASER (d) MASER
20. If the number of electrons in excited state exceeds than the number of electrons in the ground state, then this will occur: (2022)
 (a) Excitation Potential (b) Population Inversion (c) Excitation Energy (d) Ionization Energy
21. LASER produces three dimensional image called: (2022)
 (a) monogram (b) spectrogram (c) hologram (d) photogram
22. The lifetime of metastable state is of the order of: (2021)
 (a) 10^{-8}s (b) 10^{-4}s (c) **10^{-3}s** (d) 10^{-2}s
23. The principal of laser production is: (2019)
 (a) Spontaneous emission (b) Induced absorption
 (c) Spontaneous absorption (d) Stimulated emission
24. In Laser, the lifetime of an electron in metastable state is of the order of: (2021, 17, 15)
 (a) 10^{-8}s (b) 10^{-3}s (c) **10^{-3}s** (d) 10^{-1}s
25. The number of energy levels, required for laser production is: (2017)
 (a) 2 (b) 3 (c) 4 (d) 5
26. Laser is produced due to the: (2021, 16)
 (a) **Stimulated emission of radiation** (b) stimulated absorption of radiation
 (c) Spontaneous emission of radiation (d) spontaneous absorption of radiation
27. Laser produces: (2014, 12)
 (a) An electron beam (b) A neutron beam
 (c) A coherent beam of light (d) none of these
28. The laser is a device which can produce: (2001)
 (a) An intense beam of light (b) A coherent beam of light
 (c) A monochromatic beam of light (d) **All of the above**

SECTION- B: SHORT QUESTIONS

- State the basic postulates of Bohr's theory of atomic structure. Derive the expression for the radius of n^{th} orbit of a Hydrogen atom. (2021, 15)
- Derive an expression for total energy of the electron in the n^{th} orbit of a hydrogen atom. (2022)
- Derive the expressions for the wavelength of the photons emitted in a hydrogen spectrum.
- Describe the production of x-rays. Also give the characteristics and uses of x-rays.
- What is meant by LASER? Define metastable state OR population inversion. (2025)
- Describe construction and working of Helium-Neon LASER.

NUMERICAL PROBLEMS**Bohr's Atomic Model & Hydrogen Spectrum Series:**

1. Hydrogen atom in ground state is excited by absorbing a photon of 12.15eV. Find the quantum number of this excited state. (2022)
2. Light of wavelength 486.3 nm is emitted by a hydrogen atom in Balmer series, what transitions of the hydrogen atom is responsible for this radiation. (Ans: $n = 4$) (2021)
3. What is the wavelength of 3rd spectral line of Paschen series in hydrogen atom? ($R_H = 1.097 \times 10^7 \text{ m}^{-1}$). (2019)
4. A photon of what minimum energy is required to excite a hydrogen atom from $n = 1$ to $n = 3$? ($R_H = 1.097 \times 10^7 \text{ m}^{-1}$) (2018)
5. How much energy is needed to ionize a hydrogen atom originally in ground state? ($h = 6.63 \times 10^{-34} \text{ Js}$, $c = 3 \times 10^8 \text{ m/s}$, $R_H = 1.097 \times 10^7 / \text{m}$) (2017)
6. Find the shortest and the longest wavelength of emitted photons in Hydrogen spectra in Pfund series. (2016)
7. Find the shortest wavelength of photon emitted in the Balmer series and determine its energy in eV. ($R_H = 1.097 \times 10^7 \text{ m}^{-1}$) (2015, 10)
8. A photon of 12.0 eV absorbed by a hydrogen atom originally in the ground state raises the atom to an excited state. What is the quantum number of this state? (Ans: $n_f = 3$) (2013, 02E)
9. Calculate the energy of the longest wavelength radiation emitted in the Paschen series in hydrogen atom spectra. ($R_H = 1.0968 \times 10^7 \text{ m}^{-1}$, $h = 6.69 \times 10^{-34} \text{ Js}$, $C = 3 \times 10^8 \text{ m/s}$) (Ans: 0.662 eV) (2011)
10. Calculate the binding energy of Hydrogen atom. (Ans: -13.6 eV) (2007)
11. In a hydrogen atom an electron experiences transition from a state whose binding energy is 0.54eV to the state whose excitation energy is 10.2eV. ($R_H = 1.097 \times 10^7 \text{ m}^{-1}$) Calculate: (i) The quantum number of the two states (ii) The wavelength of the photon emitted. (Ans: 5, 2, $3.64 \times 10^{-7} \text{ m}$) (2005)

SECTION- C: DETAILED QUESTIONS:

12. Describe the terms optical pumping, metastable state, population inversion and stimulated emission. Write the construction and working of Helium-Neon Laser. (MP:25)